MTH102

Linear Algebra

2025–2026–I Semester

08 September 2025

Preface

This discusses the content covered in the class, along with some examples, exercises, and problem sets. You are encouraged to go through it. Solving the exercises would greatly benefit in understanding the content of the course. The typos/inaccuracies may be reported to jpsaha@iiserb.ac.in.

Timetable

The classes are scheduled on

- Mondays,
- Wednesdays,
- Fridays

from **11:00am to 11:55am** in L5, LHC. The tutorial is scheduled on

- Thursdays

from **5:30pm to 6:30pm** in L5, LHC.

Office hours

The office hours will be held in **Room no. 108, AB1** during **5:00pm to 6:00pm** on

- Mondays,
- Tuesdays,
- Wednesdays,
- Fridays.

The grading policy is as follows.

Components	Weightage
Quiz	20%
Mid-semester examination	35%
End-semester examination	45%

Calendar

The academic calendar is available at https://www.iiserb.ac.in/doaa/schedule.

The announcements related to the course will be made on the following Google classroom page.

https://classroom.google.com/u/1/c/ODAwOTg3NDg2Mjg0

The notes can be accessed at the link below (until all the students join the Google classroom).

https://jpsaha.github.io/home/blog/2025/mth102/

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Introduction

Chapter 1. Sets

Definition

A **set** is a well-defined collection of objects. The objects of a set are called its **elements** or **members**.

Examples

- The set of vowels a, e, i, o, u.
- The set consisting of $2, 4, 6, 8, 10$.
- The set consisting of $3, 15, 35, 63, 99$.

Here are further examples.

Examples

- The set of positive integers divisible by 3.
- The set of prime numbers less than 100.
- The set of positive integers which can be expressed as the product of two distinct primes.
- The set of positive integers which can be expressed as the product of two or more distinct primes.
- The set of positive integers which are smaller than 100 and share no common factor with 100.

Remark

Note that the first few sets are described by writing down all its elements, whereas in the latter examples, the sets are described by the rules or properties which determine whether a particular object is an element or not.

Notation

A set is usually denoted by an uppercase letter, for example,

$$A, B, C, X, Y, Z,$$

whereas the lowercase letters are used to denote its elements. For instance, one may use

- a to denote an element of A,
- b to denote an element of B,
- c to denote an element of C,
- x to denote an element of X,
- y to denote an element of Y,
- z to denote an element of Z.

Several special sets are denoted by certain standard symbols. Here are some such examples.

$\mathbb{N}$	the set of positive integers
$\mathbb{Z}$	the set of integers

If x is an element of a set X, then one says that **x belongs to X**, and writes

$$x \in X.$$

Note that 2 belongs to $\mathbb{N}$, but $\frac{1}{2}$ does not belong to $\mathbb{N}$. One writes

$$2 \in \mathbb{N}, \frac{1}{2} \notin \mathbb{N}.$$

Notation

There are two ways to write down a set. For example, consider the set of vowels, which is written as

$$V = \{\text{a, e, i, o, u}\}.$$

Note that the elements are seperated by commas and enclosed in braces $\{\}$.

Another way to write down a set is by stating the *rules* or *properties* which determine whether a particular object is an element or not. For example, the set of even positive integers is written as

$$E = \{x \in \mathbb{Z} : x \text{ is even}, x > 0\}.$$

This reads

"E is the set of elements x in $\mathbb{Z}$ such that x is even and $x > 0$".

Example

Note that the set $A = \{3, 15, 35, 63, 99\}$ is equal to

$$\{x \in \mathbb{Z} : x \text{ is equal to } n^2 - 1 \text{ for some } n \in \{2, 4, 6, 8, 10\}\}.$$

Since $1, 2$ are the roots of the polynomial $x^2 - 3x + 2$, it follows that

$$\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}.$$

Also note that

$$\mathbb{N} = \{1, 2, 3, \ldots\},$$
$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \ldots\}.$$

One uses the symbol $:=$ to indicate that the symbol on the left is being defined by the symbol on the right.

$$\mathbb{Q} := \left\{\frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0\right\}.$$

It is called the *set of rational numbers*.

If the number of elements of a set is finite, then it is called a **finite set**. If a set is not finite, it is called an **infinite set**.

Example

The sets

$$\{\text{a, e, i, o, u}\}, \{3, 15, 35, 63, 99\}, \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$$

are finite. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are infinite.

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Solution. Note that 1 is the only element of $\mathbb{N}$ satisfying the equation $x^2 - 1 = 0$. This gives

$$\{x \in \mathbb{N} : x^2 - 1 = 0\} = \{1\}.$$

Since $1, -1$ are precisely all the elements of $\mathbb{Z}$ satisfying $x^2 - 1 = 0$, it follows that

$$\{x \in \mathbb{Z} : x^2 - 1 = 0\} = \{1, -1\}.$$

□

§1.1 Basic terminologies

If A, B are sets, and every element of A also belongs to B, then we say that A is a **subset** of B, and write

$$A \subseteq B.$$

If A is not a subset of B, one writes

$$A \not\subseteq B.$$

Example

- Note that every set is a subset of itself.
- Note that $\mathbb{N}$ is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is a subset of $\mathbb{Q}$. One writes

$$\mathbb{N} \subseteq \mathbb{Z}, \quad \mathbb{Z} \subseteq \mathbb{Q}.$$

- Also note that

$$\mathbb{Z} \not\subseteq \mathbb{N}, \mathbb{Q} \not\subseteq \mathbb{Z}, \mathbb{Q} \not\subseteq \mathbb{N}.$$

If A is a subset of B and A is not equal to B, then A is said to be a *proper subset of* B, and one writes

$$A \subsetneq B.$$

Note that

$$\mathbb{N} \subsetneq \mathbb{Z}, \mathbb{Z} \subsetneq \mathbb{Q}.$$

Exercise 1.2. What does it mean to say that A is not a subset of B?

Solution. Note that A is a subset of B is equivalent to the statement that each element of A lies in B. Hence, A is not a subset of B is equivalent to saying that some element of A does not belong to B. □

Exercise 1.3. Show that if A is a subset of B and B is a subset of C, then A is a subset of C.

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

🔥 **Tip**

Try to show that no natural number n satisfies any of the equations

$$n^2 = 2, n^2 = 3.$$

Consider the set

$$A = \{n \in \mathbb{N} : n^4 - 5n^2 + 6 = 0\}.$$

Note that the set A has no elements. Any such set is called the **empty set** or **null set**, and is denoted by the symbol

$$\emptyset.$$

Two sets, A and B, are said to be **equal**[1] if any element of A belongs to B, and any element of B also belongs to A, or equivalently,

$$A \subseteq B \text{ and } B \subseteq A$$

holds. If two sets A, B are not equal, one writes

$$A \neq B.$$

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

🔥 **Tip**

Does Exercise 1.2 help?

§1.2 Operations on sets

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

[1] *It* may appear obvious! The advantage of putting forth definitions is to set up notations and conventions, so that these are not left to interpretations!

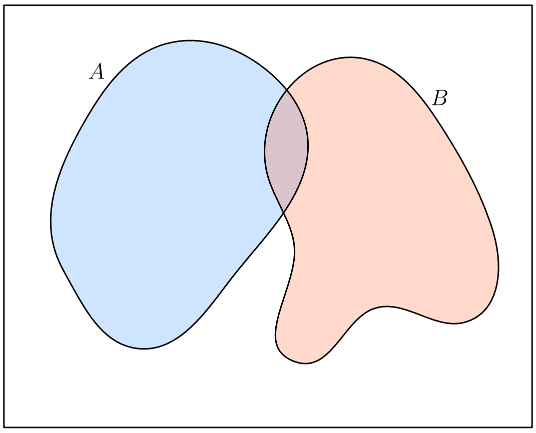

Figure 1: Two sets A and B

Definition

If A, B are sets, their **union** is denoted by $A \cup B$, which is defined as

$$A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

See Figure 2.

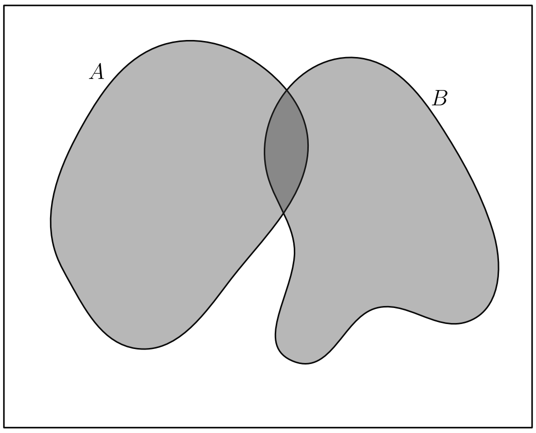

Figure 2: The union of A and B

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Usage of "or" in the inclusive sense

The word "or" is used in the inclusive sense, allowing two or more of the conditions to be satisfied. In common terminology, this inclusive sense is denoted by "and/or".

Exercise 1.8. If A is a set, show that

$$A \cup A = A.$$

Exercise 1.9 (Commutative property)**.** If A, B are sets, show that

$$A \cup B = B \cup A.$$

Definition

If A, B, C are sets, their **union** is denoted by $A \cup B \cup C$, which is defined as

$$A \cup B \cup C := \{x : x \in A \text{ or } x \in B \text{ or } x \in C\}.$$

Exercise 1.10 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Use Exercise 1.9 and Exercise 1.10, to show that for any three sets A, B, C, the following sets are equal

$$A \cup B \cup C, A \cup C \cup B, B \cup A \cup C, B \cup C \cup A, C \cup A \cup B, C \cup B \cup A.$$

Definition

If A, B are sets, their **intersection** is denoted by $A \cap B$, which is defined as

$$A \cap B := \{x : x \in A \text{ and } x \in B\}.$$

See Figure 3.

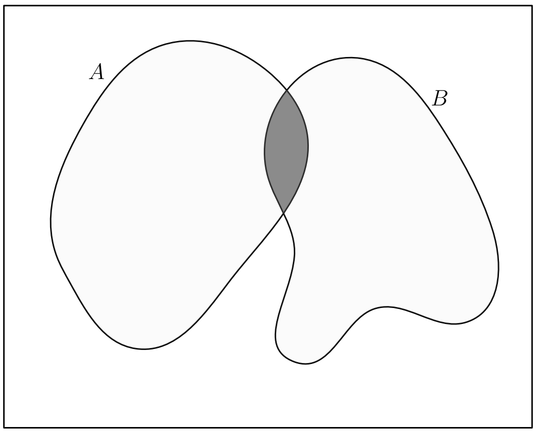

Figure 3: The intersection of A and B

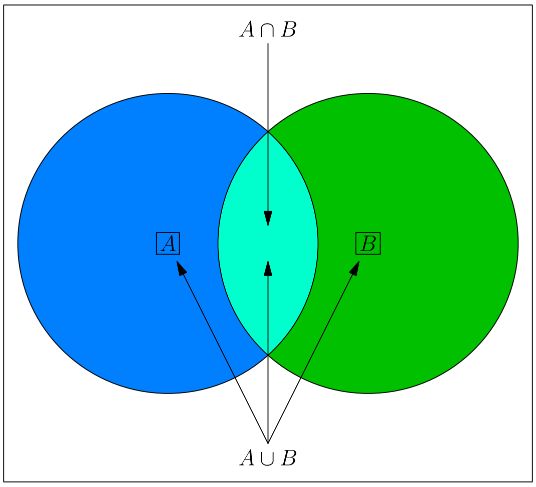

Figure 4: The union and intersection of A and B

Exercise 1.11. Identify the integers among

$$1, 2, 3, 4, \ldots, 20$$

which are divisible by at least one of 2 and 5.

Lemma 1.12. *Let X be a set, and A, B be finite subsets of X. Then*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Proof. Note that the set $A \cup B$ is equal to the union of the disjoint sets A and $B \setminus A$. This shows that

$$|A \cup B| = |A| + |B \setminus A|.$$

Also note that the set B is equal to the union of the disjoint subsets $A \cap B$ and $B \setminus A$. This gives

$$|B| = |A \cap B| + |B \setminus A|.$$

It follows that

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

□

Exercise 1.13. Using Lemma 1.12 or otherwise, show that for finite subsets A, B, C of a set X,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Solution. Note that

$$
\begin{aligned}
&|A \cup B \cup C| \\
&= |(A \cup B) \cup C| \\
&= |A \cup B| + |C| - |(A \cup B) \cap C| && \text{(by Lemma 1.12)} \\
&= |A \cup B| + |C| - |(A \cap C) \cup (B \cap C)| && \text{(by Exercise 1.17)} \\
&= |A \cup B| + |C| - (|A \cap C| + |B \cap C| - |(A \cap C) \cap (B \cap C)| && \text{(by Lemma 1.12)} \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |(A \cap C) \cap (B \cap C)| \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \\
&= |A| + |B| - |A \cap B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| && \text{(by Lemma 1.12)} \\
&= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.
\end{aligned}
$$

□

Exercise 1.14. If A is a set, show that

$$A \cap A = A.$$

Exercise 1.15 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Definition

If A, B, C are sets, their **intersection** is denoted by $A \cap B \cap C$, which is defined as

$$A \cap B \cap C := \{x : x \in A \text{ and } x \in B \text{ and } x \in C\}.$$

Exercise 1.16 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Exercise 1.17 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Tip

Recall when two sets are equal.

Remark

All sets under consideration in a given context, are assumed to be contained in a 'large' fixed set, called the *universal set*. For example, while studying the positive integers (for instance, some sets consisting of the positive integers), the set $\mathbb{N}$ can be taken as the universal set. While studying the integers, the set $\mathbb{Z}$ can be taken as the universal set.

Definition

Let A, B be subsets of X. The **complement of A in B** is denoted by $B \setminus A$, and is defined by

$$B \setminus A := \{x \in X : x \in B, x \notin A\}$$

It is also called the **difference of B and A**.

The complement of A in X, is called the *complement of* A, and is denoted by A^c, and is defined to be

$$A^c := \{x \in X : x \notin A\}.$$

Exercise 1.18. Show that

$$A^c = X \setminus A,$$

where X denotes the underlying universal set.

Exercise 1.19. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.20. If A, B are subsets of X, show that

$$A^c \cap B = B \setminus A.$$

Exercise 1.21. If A is a subset of X, then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.22. If A, B are subsets of X, then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Definition

Two sets A, B are said to be **disjoint** if

$$A \cap B = \emptyset.$$

Exercise 1.23. Show that for any sets A, B, the following inclusions hold.

$$A \cap B \subseteq A,$$
$$A \cap B \subseteq B,$$
$$A \subseteq A \cup B,$$
$$B \subseteq A \cup B.$$

Exercise 1.24. Show that for any sets A, B, the following are equivalent.

$$A \subseteq B,$$
$$A \cap B = A,$$
$$A \cup B = B,$$
$$B^c \subseteq A^c.$$

Exercise 1.25. If A, B are sets, show that
1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$,
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Definition

The **symmetric difference** of two sets A, B, denoted by $A \Delta B$, is defined as

$$A \Delta B := (A \cup B) \setminus (A \cap B).$$

Exercise 1.26. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.27. If A, B are sets, then show that

$$A \Delta B = B \Delta A,$$
$$A \cup B = (A \Delta B) \cup (A \cap B),$$

and

$$(A \Delta B) \cap (A \cap B) = \emptyset.$$

Exercise 1.28. Do there exist three finite sets A, B, C satisfying

$$|A \Delta B| = 1, |B \Delta C| = 2, |C \Delta A| = 4?$$

§1.3 Family of sets

Definition

Let $n \geq 2$ be an integer, and let $A_1, \ldots, A_n$ be sets. The **union of $A_1, \ldots, A_n$** is denoted by $A_1 \cup \cdots \cup A_n$, and is defined by

$$A_1 \cup \cdots \cup A_n := \{x : x \text{ belongs to } A_i \text{ for some } 1 \leq i \leq n\}.$$

The **intersection of $A_1, \ldots, A_n$** is denoted by $A_1 \cap \cdots \cap A_n$, and is defined by

$$A_1 \cap \cdots \cap A_n := \{x : x \text{ belongs to } A_i \text{ for all } 1 \leq i \leq n\}.$$

§1.4 Power set

Example

Note that the subsets of $\{1, 2, 3\}$ are

$$\begin{gathered} \{1,2,3\}, \\ \{1,2\}, \{2,3\}, \{3,1\}, \\ \{1\}, \{2\}, \{3\}, \\ \emptyset. \end{gathered}$$

Definition

Let A be a set. The set of subsets of A is denoted by $\mathcal{P}(A)$. It is called the **power set of A**.

Example

Note that

$$\begin{aligned} \mathcal{P}(\{1\}) &= \{\emptyset, \{1\}\}, \\ \mathcal{P}(\{x\}) &= \{\emptyset, \{x\}\}, \\ \mathcal{P}(\{x,y\}) &= \{\emptyset, \{x\}, \{y\}, \{x,y\}\}, \\ \mathcal{P}(\{x,y,z\}) &= \{\emptyset, \{x\}, \{y\}, \{z\}, \{x,y\}, \{y,z\}, \{z,x\}, \{x,y,z\}\}, \\ \mathcal{P}(\{1,2,3\}) &= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}, \\ \mathcal{P}(\{1,2,3,{\color{red}4}\}) &= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}, \\ &\quad \{{\color{red}4}\}, \{1,{\color{red}4}\}, \{2,{\color{red}4}\}, \{3,{\color{red}4}\}, \{1,2,{\color{red}4}\}, \{2,3,{\color{red}4}\}, \{1,3,{\color{red}4}\}, \{1,2,3,{\color{red}4}\}\}. \end{aligned}$$

Remark

Convince yourself that for any integer $n \geq 0$, the power set of a set of size n has size 2^n.

§1.5 Cartesian product of sets

Definition

If A, B are sets, then the **cartesian product** $A \times B$ of A and B is the set of all ordered pairs of the form (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

If $A = \{1, 2\}$ and $B = \{2, 3, 4\}$, then

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4)\}.$$

Exercise 1.29. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X, its number of elements is denoted by $|X|$.

Determine the cartesian product of two suitable subsets of $\mathbb{R}$.

§1.6 Sets and subsets of real and complex numbers

$$\begin{aligned}
\mathbb{N} &= \text{the set of positive integers} \\
&= \{1, 2, 3, \dots\}, \\
\mathbb{Z} &= \text{the set of integers} \\
&= \{0, 1, -1, 2, -2, 3, -3, \dots\}, \\
\mathbb{Q} &= \text{the set of rational numbers} \\
&= \left\{\frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N}\right\}, \\
\mathbb{R} &= \text{the set of real numbers.}
\end{aligned}$$

Remark

There are real numbers which are not rational. For instance, the number $\sqrt{2}$ is not rational.

Lemma 1.30. *No rational number x satisfies $x^2 = 2$.*

Proof. On the contrary, let us assume that some rational number x satisfies $x^2 = 2$. Replacing x by $-x$ if necessary, we may and do assume that x is positive. Write $x = \frac{a}{b}$ for some positive integers a, b. Let d denote the greatest common divisor of a, b. Write $a = da_0, b = db_0$ where a_0, b_0 are positive integers. Note that $\left(\frac{a_0}{b_0}\right)^2 = 2$.

Claim

None of a_0, b_0 is divisible by 3.

Proof of the Claim

On the contrary, let us assume that 3 divides at least one of the integers a_0, b_0.

If 3 divides a_0, then 3 divides a_0^2. Using $a_0^2 = 2b_0^2$, it follows that 3 divides b_0. Further, if 3 divides b_0, then 3 divides $2b_0^2$, and hence 3 divides a_0^2. This shows that a_0 is divisible by 3.

In both the cases, it follows that the greatest common divisor of a_0, b_0 is larger than 1. This shows that the greatest common divisor of da_0, db_0 is larger than d. Using $a = da_0, b = db_0$, we obtain that the greatest common divisor of a, b is larger than d, which is impossible. This proves that the assumption that some rational number x satisfies $x^2 = 2$ is not tenable. This proves the claim.

Claim

If n is an integer not divisible by 3, then n^2 is equal to $3k + 1$ for some integer k, depending on n.

Proof of the Claim

Let q (resp. r) denote the quotient (resp. remainder) obtained upon dividing n by 3. This gives $n = 3q + r$. Note that

$$\begin{aligned} n^2 &= (3q + r)^2 \\ &= 9q^2 + 6qr + r^2 \end{aligned}$$

Since $n = 3q + r$ and 3 does not divide n, it follows that r is equal to 1 or 2. If $r = 1$, then

$$n^2 = 3(3q^2 + 2qr) + 1.$$

If $r = 2$, then

$$n^2 = 3(3q^2 + 2qr) + 4 = 3(3q^2 + 2qr + 1) + 1.$$

This proves the claim.

Using the claims above, it follows that

$$\begin{aligned} a_0^2 &= 3c + 1, \\ b_0^2 &= 3d + 1 \end{aligned}$$

for some integers c, d. This gives

$$3c + 1 = 2(3d + 1).$$

This yields $1 = 3(c - 2d)$, which is impossible.

This completes the proof of the lemma. $\square$

Note that no real number x satisfies $x^2 = -1$. To "resolve" this, one "introduces"[2] an element i satisfying

$$\boxed{i^2 = -1.}$$

[2]To introduce is to adjoin.

The element i can be multiplied by any real number, and the product can also be added to any real number. This leads to the numbers of the form $a + ib$, where a, b are real numbers. Such numbers are called the **complex numbers**, and the set of such numbers is denoted by $\mathbb{C}$.

Definition

The **set of complex numbers**, is denoted by $\mathbb{C}$, and is defined by

$$\mathbb{C} := \{a + ib : a, b \in \mathbb{R}\}.$$

More formally, one defines $\mathbb{C}$, as the following set

$$\{(a, b) : a, b \in \mathbb{R}\},$$

equipped with addition and multiplication defined by

$$\begin{aligned}(a, b) + (c, d) &:= (a + c, b + d),\\ (a, b) \cdot (c, d) &:= (ac - bd, ad + bc),\end{aligned}$$

for any $(a, b), (c, d)$ with $a, b, c, d \in \mathbb{R}$.

Chapter 2. Induction principles

Example

Show that

$$\begin{aligned} 1+2+3+\cdots+n &= \frac{n(n+1)}{2}, \\ 1^2+2^2+3^2+\cdots+n^2 &= \frac{n(n+1)(2n+1)}{6}, \\ 1^3+2^3+3^3+\cdots+n^3 &= \left(\frac{n(n+1)}{2}\right)^2 \end{aligned}$$

hold for any positive integer n.

Solution

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$1+2+3+\cdots+n = \frac{n(n+1)}{2}.$$

Note that $P(1)$ holds. Let k be an element of $\mathbb{N}$, and assume that $P(k)$ holds. Note that

$$\begin{aligned} 1+2+3+\cdots+k+(k+1) &= \frac{k(k+1)}{2}+(k+1) \quad (\text{using } P(k)) \\ &= (k+1)\left(\frac{k}{2}+1\right) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n.

If A is a nonempty subset of $\mathbb{N}$, then an element a_0 of A is called a **least element** of A if $a_0 \leq a$ for all $a \in A$.

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A, then show that $a_1 = a_2$.

Remark

By Exercise 2.1, a nonempty subset of $\mathbb{N}$ has at most one least element, to be called **the** least element, if it exists.

Well-ordering principle

Every nonempty subset of $\mathbb{N}$ has a least element.

Principle of mathematical induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(n)$ holds for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

Principle of strong induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(1), P(2), ..., P(n)$ hold for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

i **Remark**

The above three principles are equivalent.

Exercise 2.2 (Infinitude of primes). Let $a_1 = 2$, and $a_{n+1} = a_n(a_n + 1)$ for any positive integer n. For any $n \in \mathbb{N}$, show that a_n has at least n distinct prime divisors.

Solution. For a positive integer n, let $P(n)$ denote the statement that $a_n \geq 1$ and a_n has at least n distinct prime divisors.

Since $a_1 = 2$, it follows that $P(1)$ is true.

Let n be a positive integer such that $P(n)$ holds. This gives that $a_n \geq 1$, and hence $a_n + 1 \geq 2$. Also note that the integers $a_n, a_n + 1$ have no common prime divisor, since any of their common divisors divides their difference, which is equal to 1. Hence, no prime divisor of a_n is a prime divisor of $a_n + 1$. By the induction hypothesis, a_n has at least n distinct prime divisors. Since $a_n + 1 \geq 2$, it admits at least one prime divisor. It follows that the product $a_n(a_n + 1)$ has at least $n + 1$ distinct prime divisors. Also note that $a_n(a_n + 1) \geq 2$ holds. This shows that $P(n + 1)$ holds.

By induction, $P(n)$ holds for all $n \in \mathbb{N}$. In particular, a_n has at least n distinct prime divisors. □

Exercise 2.3. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.4. Show that

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 2.5. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 2.6. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1} n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.7. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 2.8. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 2.9. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.10. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Exercise 2.11. Let the integers $x_1, x_2, \ldots$ be defined by

$$\begin{aligned} x_1 &:= 1, \\ x_2 &:= 2, \\ x_{n+2} &:= \frac{1}{2}(x_n + n_{n+1}) \end{aligned}$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.